UF | College of Medicine
UNIVERSITY of FLORIDA

fusion

Functional Unit State Identification & Navigation with WSI QUANTIFIER

FUSION: AI-Powered Spatial Multi-Omics x Histopathology Analysis Platform

High-resolution histopathology + spatial omics + machine learning in one platform

ISMB 2026 • FUSION Demo Session

University of Florida
Computational Microscopy Imaging Laboratory
Funding: NIH U54DK134301 and OT2OD033753 (HuBMAP)

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CMIL Computational Microscopy Imaging Laboratory

SESSION ROADMAP

FUSION Demo Session

01 Why Fuse Histology + Spatial Omics

02 What is Fusion? How it is different compared to other OMICS tools?

03 The FUSION Analysis Pipeline and Supported Data Type

04 Live Fusion Walkthrough via Fusion JupyterHub Interface

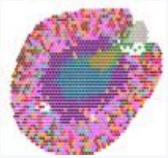
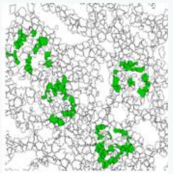
05 Heading Next: FUSION 2.0

THE INTEGRATION GAP

Why Fuse Histology and Spatial Omics?

Omics tells you what molecules are present. Histology tells you the morphological properties of Functional Tissue Units (FTU).

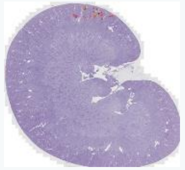
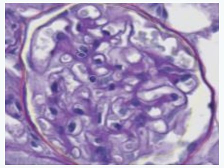
Objective: Combined processing of both modalities and perform analysis, per FTU.

Spatial Omics

Molecular readout

- Transcript counts per spot / cell
- Reference-atlas deconvolution
- Cell-type proportions

Histopathology

Structural readout

- FTU shape & basic morphology
- Quantitative pathomic features
- Gold-standard clinical context

Machine Learning

The bridge

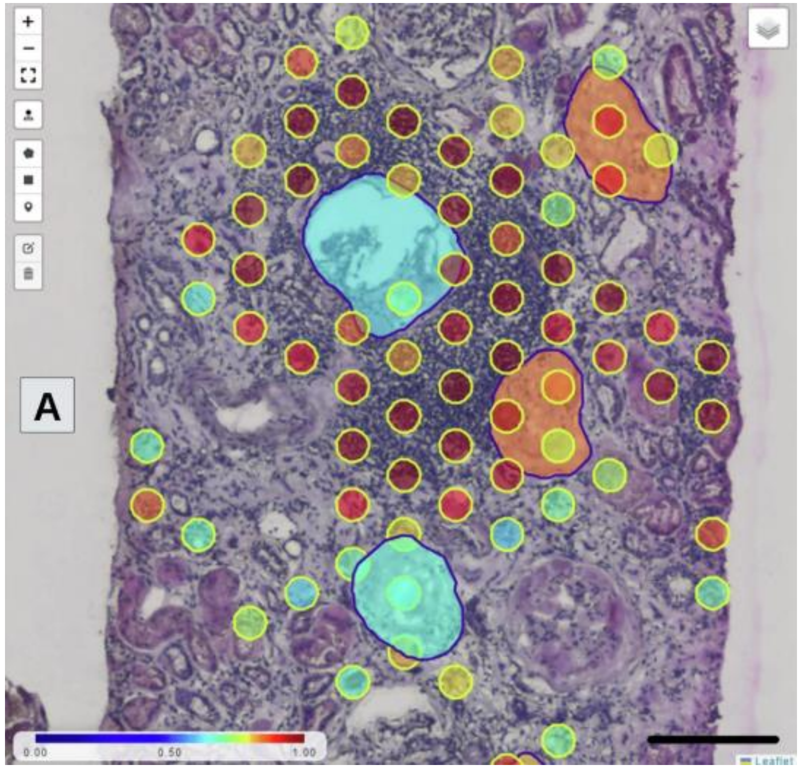
- Deep-learning FTU segmentation
- Overlap-weighted spatial aggregation

FUSION's mission: combined record per FTU— morphology + molecular composition.

PLATFORM CONCEPT

What Is FUSION: The Core Idea

A one-stop shop for typically siloed analyses - morphology and molecular biology of FTUs, in one platform.



High-Resolution AI-Based Histopathology + Spatial Omics

- Anchored to segmented structures, not arbitrary bins/grids
- Morphology and omics in a joint record and unified processing
- Every processing step: a reusable, versioned plugin
- Executes in cloud-HPC (no user infra set up needed)
- Availability of open-sourced pre-existing data or ingest user data

Image sources: Wu et al. 2024; Border et al. 2025

Let's Get STARTED

What Do You Need to Know to Run FUSION?

Whole Slide Image

H&E or PAS stain. Frozen or FFPE tissue, high resolution. 40X magnification and 0.25 MPP preferred.

+

Paired Spatial Omics Assay

Seurat v5 (.RDS) or Scanpy (.h5ad) — same tissue section. (Not direct sequencing input like raw FASTQs)

MCS

Multi-Compartment Segmentation

Segments histopathological FTU compartments

PFE

FTU Pathomic Feature Extraction

Extracts morphology per FTU

Cell Type Annotation

Label Transfer + Spot Annotation

Cell-type proportions / spot

FTU Aggregation

FTU-Level Molecular Aggregation

Spatio-molecular composition mapped to individual FTUs

Visual Analytics

Visualization & Analysis

Interactive plots, lasso select, custom analysis code execution

A flexible plugin architecture: raw WSI + molecular profiles -> a structure-aware, multi-omic feature set.

Multi-Compartment Segmentation (MCS): The Anchor

MCS → PFE → Cell Type Annot. → FTU Aggregation → Visual Analytics

- Deep-learning based AI auto-segmentation: trained model across diverse kidney disease types and stains
- Fully automated analysis pipeline — no manual annotation required

INPUT

High-resolution H&E- or PAS-stained WSI

OUTPUT

Segmented FTU structures with annotation boundaries written in a downloadable JSON file

AI generated renal FTU segmentation in high magnification. a: Non-Globally Sclerotic Glomerulus; b: Globally Sclerotic Glomerulus; c: Cortical Interstitial space; d: Cortical (green) and Medullary (blue) areas; e: Artery: f: Cortical tubules; g: Peritubular Capillaries; h: Podocyte nuclei.

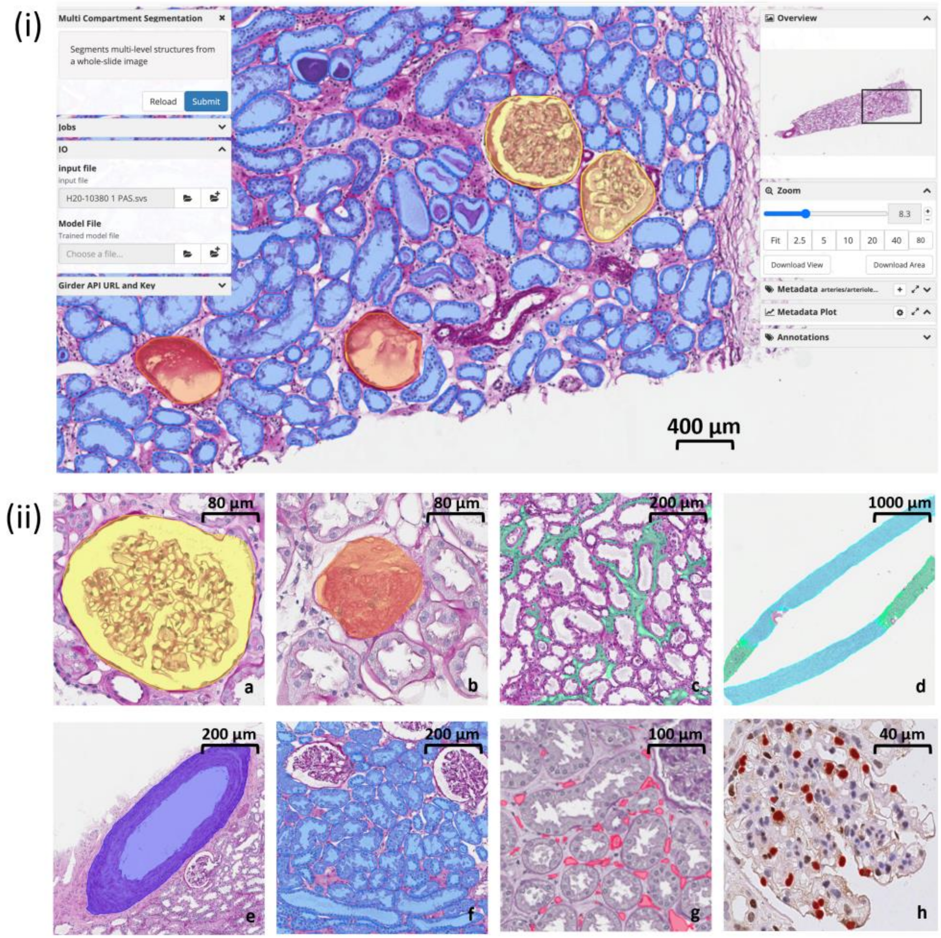
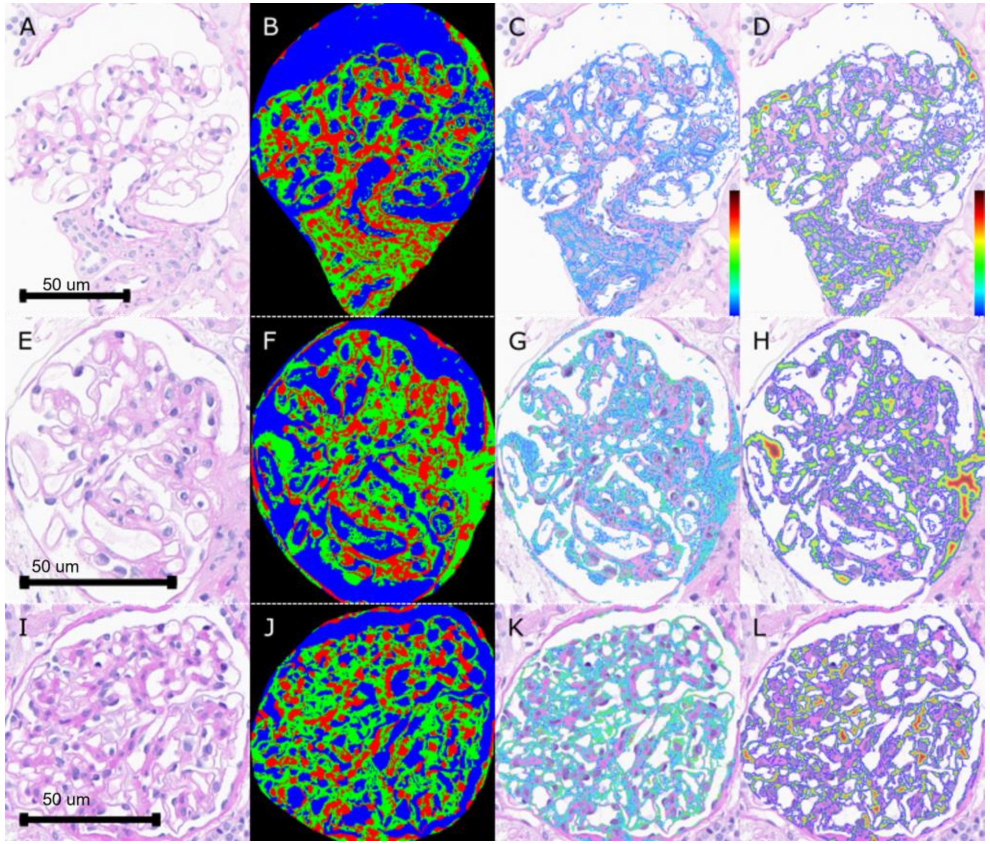


Image sources: Sarder et al. 2026

FTU Pathomic Feature Extraction (PFE)

MCS → **PFE** → Cell Type Annot. → FTU Aggregation → Visual Analytics



- FTU-aware 72 features: morphology, texture, color, spatio-nuclei organization
- Computed across 3 sub-compartments per FTU: nuclei, PAS+ (cytoplasm/matrix), and lumen

INPUT

High-resolution H&E-stained WSI with segmented FTUs

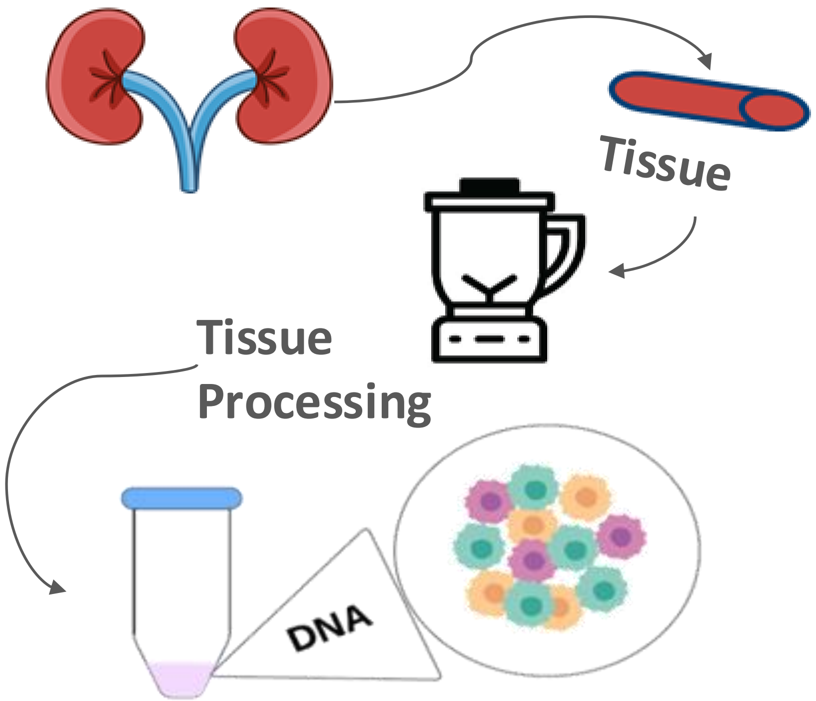
OUTPUT

Per-FTU pathomic feature table in csv (72 features), downloadable, exportable and linkable to omics data

Segmented (A, E, F) and sub-compartmentalized (B, F, J: nuclei red, PAS+ green, lumen blue) glomerular masks from different stages of disease progression: (C, G, K) standard deviation of PAS+ color mask, smoother color differences between glomerular capillary and lumen indicative of a normal healthy glomerulus. (D, H, L) mapped distance transform indicating fibrotic/sclerotic pathways inside glomerulus with greater PAS+ thickness.

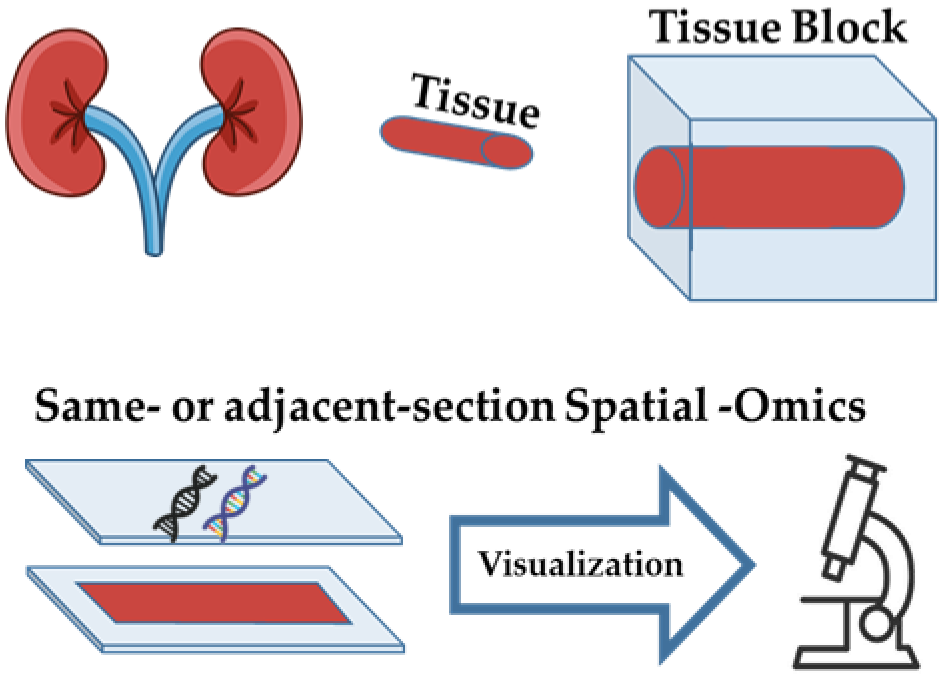
Image sources: Sarder et al. 2026

Background:
Bulk and Single Cell Omics



- Learn molecular information from dissociated tissues
- Loss of the spatial dimension
- Cellular neighborhoods NOT preserved

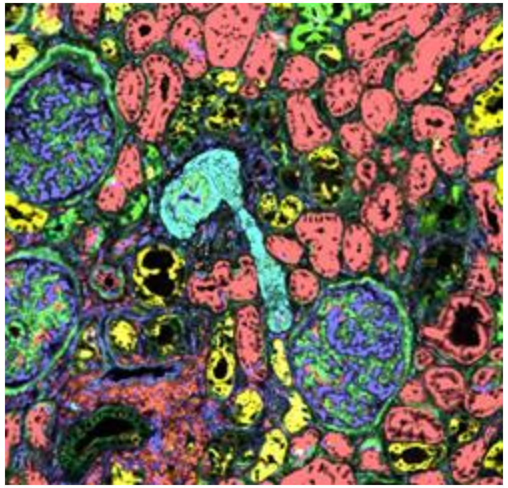
Spatial Omics



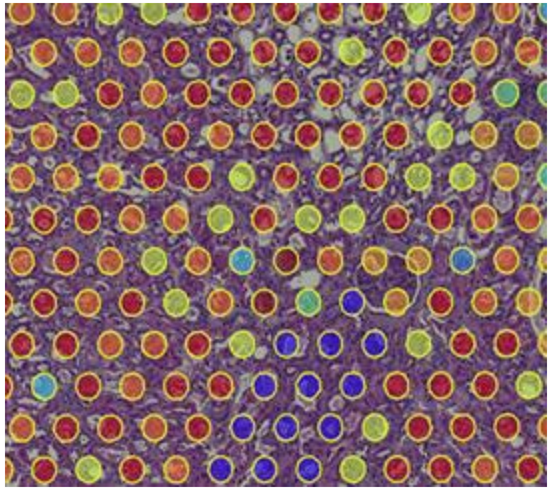
- Measure molecular information from in-tact tissue
- Spatial dimension added to data
- Preserves cellular neighborhoods

OMICS DATA TYPE DEEP DIVE

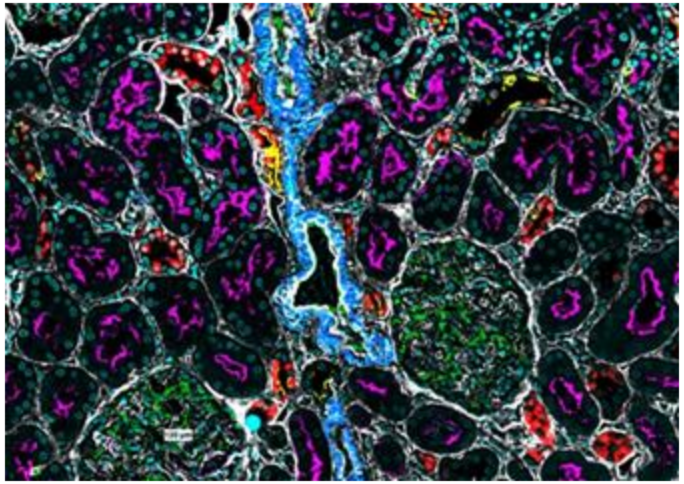
Spatial Omics: What’s Underneath the Surface?



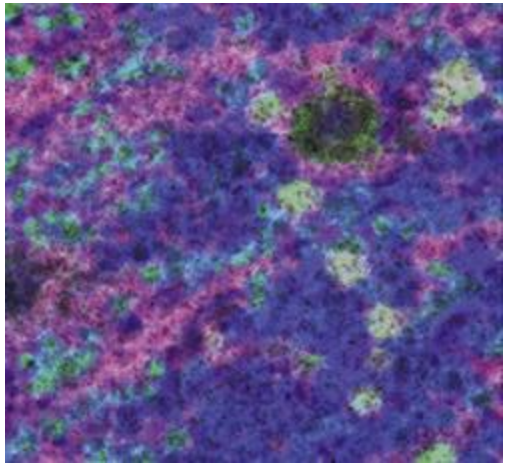
Phenocycler, A.K.A. CODEX



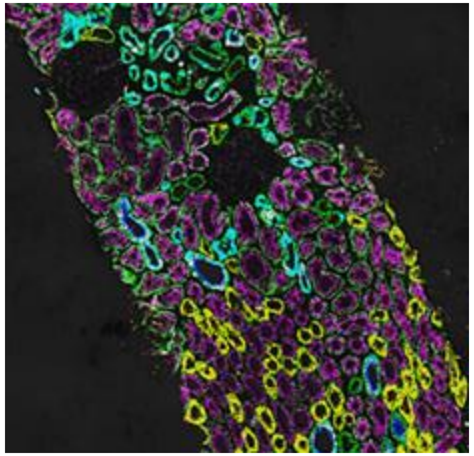
10x Visium



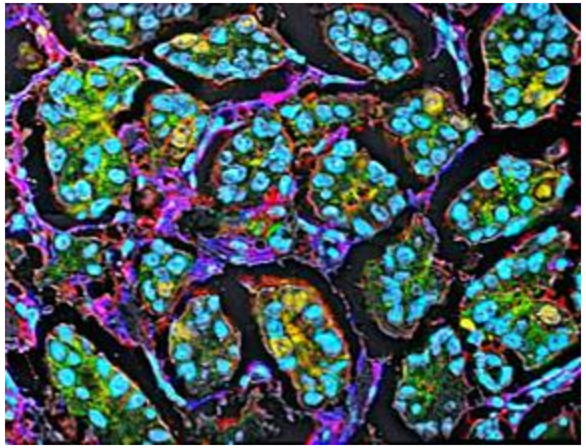
miFISH



Imaging Mass Spectrometry



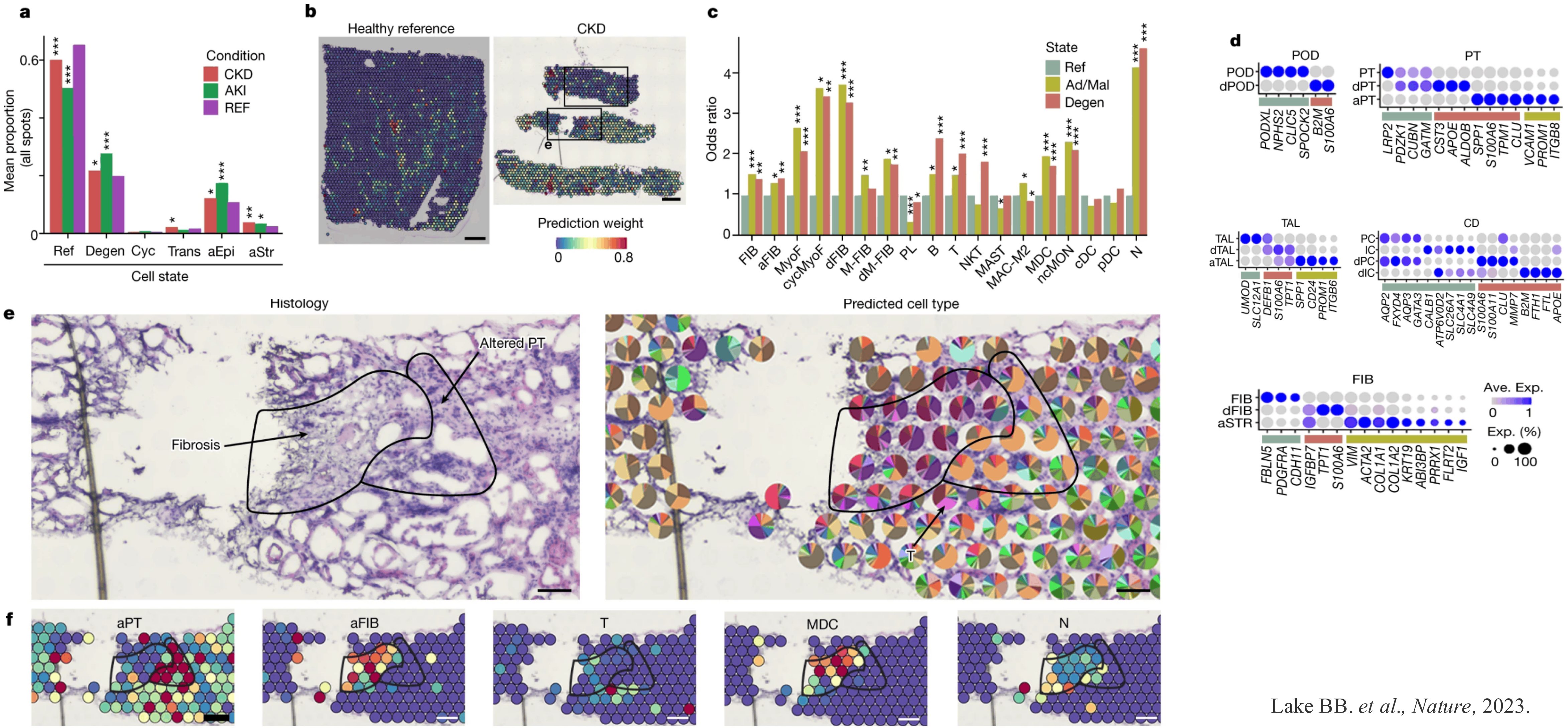
Imaging Mass Cytometry



Multiplexed Ion Beam Imaging

OMICS DATA TYPE DEEP DIVE

Omics Studies

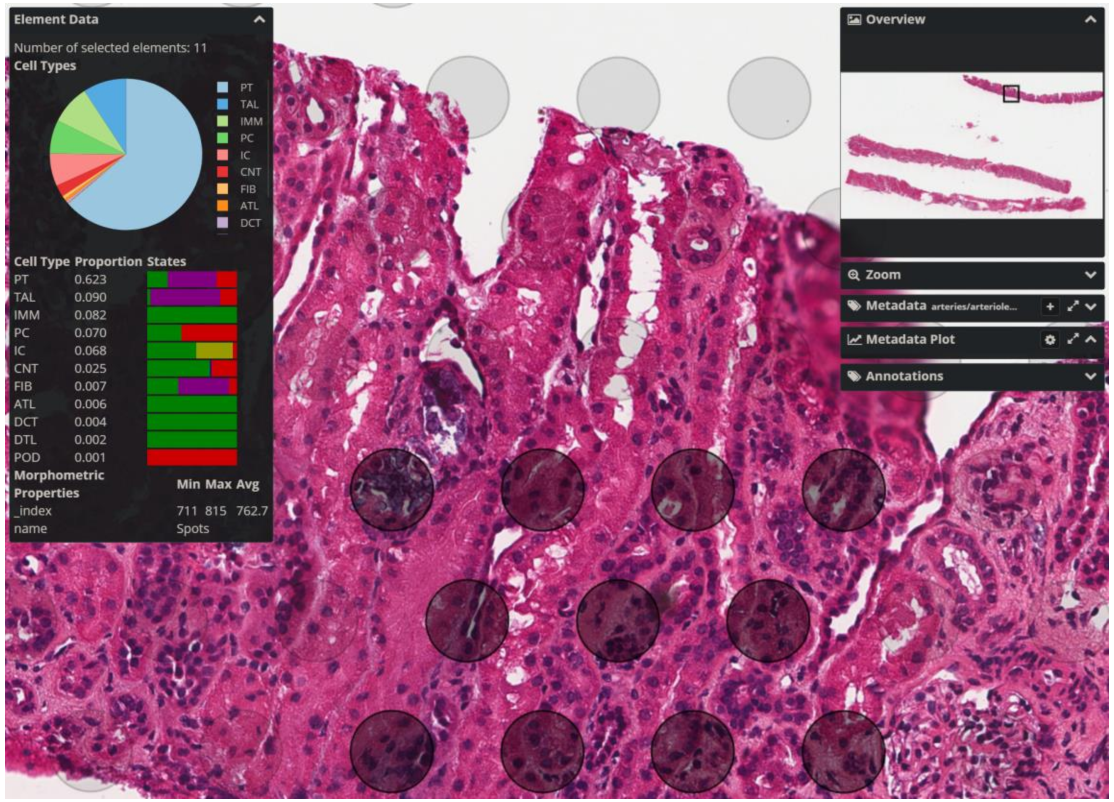


Lake BB. *et al., Nature,* 2023.

Cell Type Annotation: Label Transfer + Spot Annotation

MCS → PFE → **Cell Type Annot.** → FTU Aggregation → Visual Analytics

- Label Transfer via Seurat's anchor-based TransferData against a reference (default KPMP kidney reference)
- Deconvolves 55 µm gene expression Visium spots into predicted cell-type proportions



INPUT

Omics feature counts (Seuratrds/Scanpy h5ad) + a cell type reference(e.g. KPMP Kidney Reference)

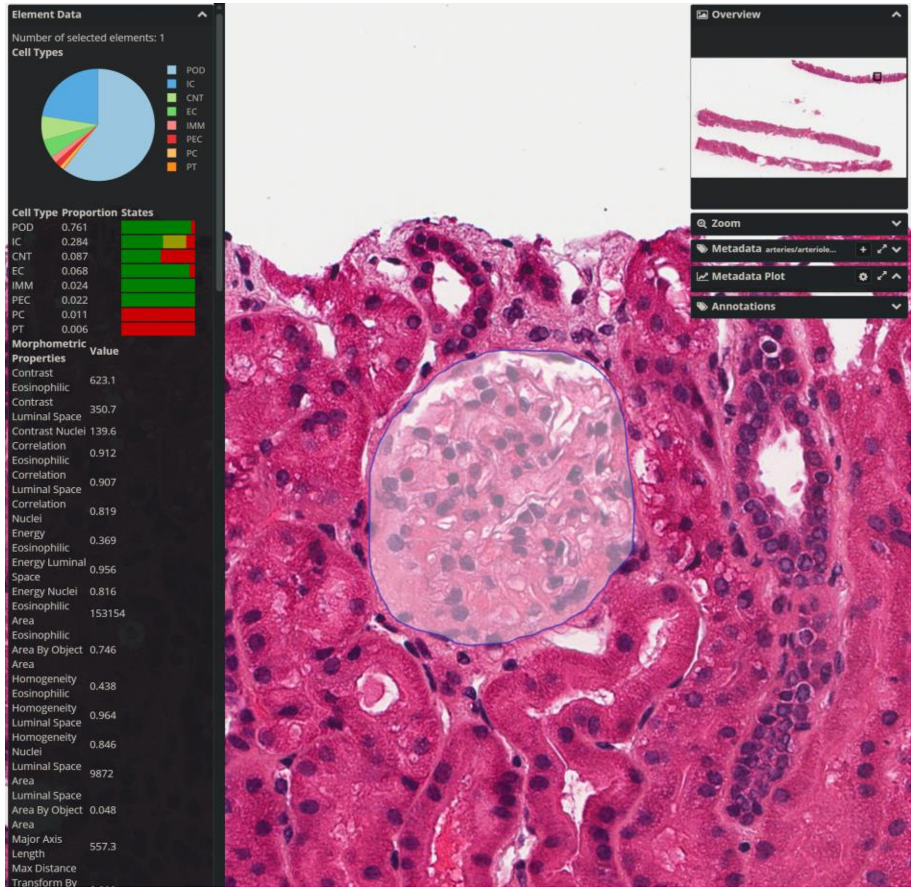
OUTPUT

Every spot annotated with cell type proportions in a downloadable JSON file

Functional Tissue Unit (FTU) Aggregation of Cell Type Annotations

MCS → PFE → Cell Type Annot. → **FTU Aggregation** → Visual Analytics

- Overlap-weighted sum of cell-type proportions of all spots at each FTU (Partial spot overlap counts proportionally)
- Merged FTU morphological and molecular signatures integrated in a unified UI for hypothesis generation



INPUT

Annotated spots + AI-segmented FTU boundaries (JSONs) + FTU Pathomics (csv)

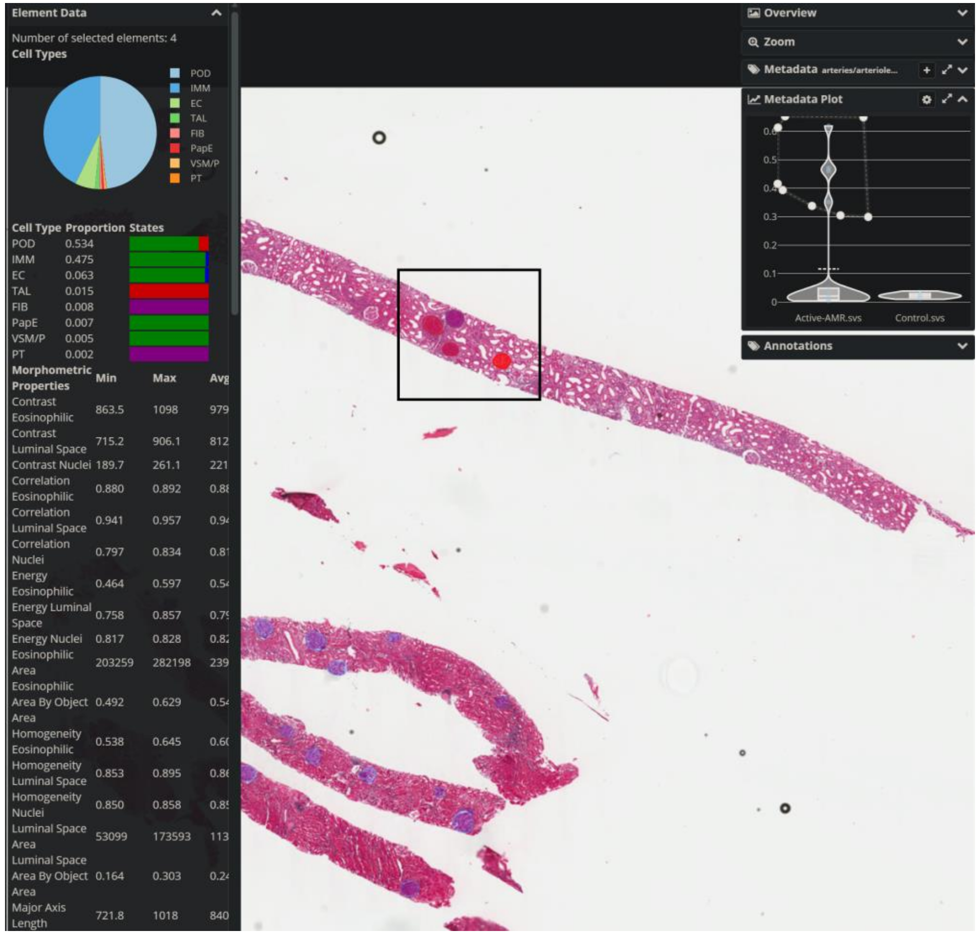
OUTPUT

Per-FTU cell-type composition joined with pathomic morphometrics (single integrated JSON file)

Visualizing Morphology + Omics in a Unified Interface

MCS → PFE → Cell Type Annot. → FTU Aggregation → **Visual Analytics**

Interactive in-tool visualizations to turn FTU-level cell-type composition and pathomic records into discovery



Select & Summarize

Rectangle/lasso-select FTUs or spots on the WSI - live statistical summary of cell-type proportions and pathomic features

Scatter Plots

Correlate any pathomic feature against another, or against cell-type proportions, to surface morphology-omics relationships

Violin Plots

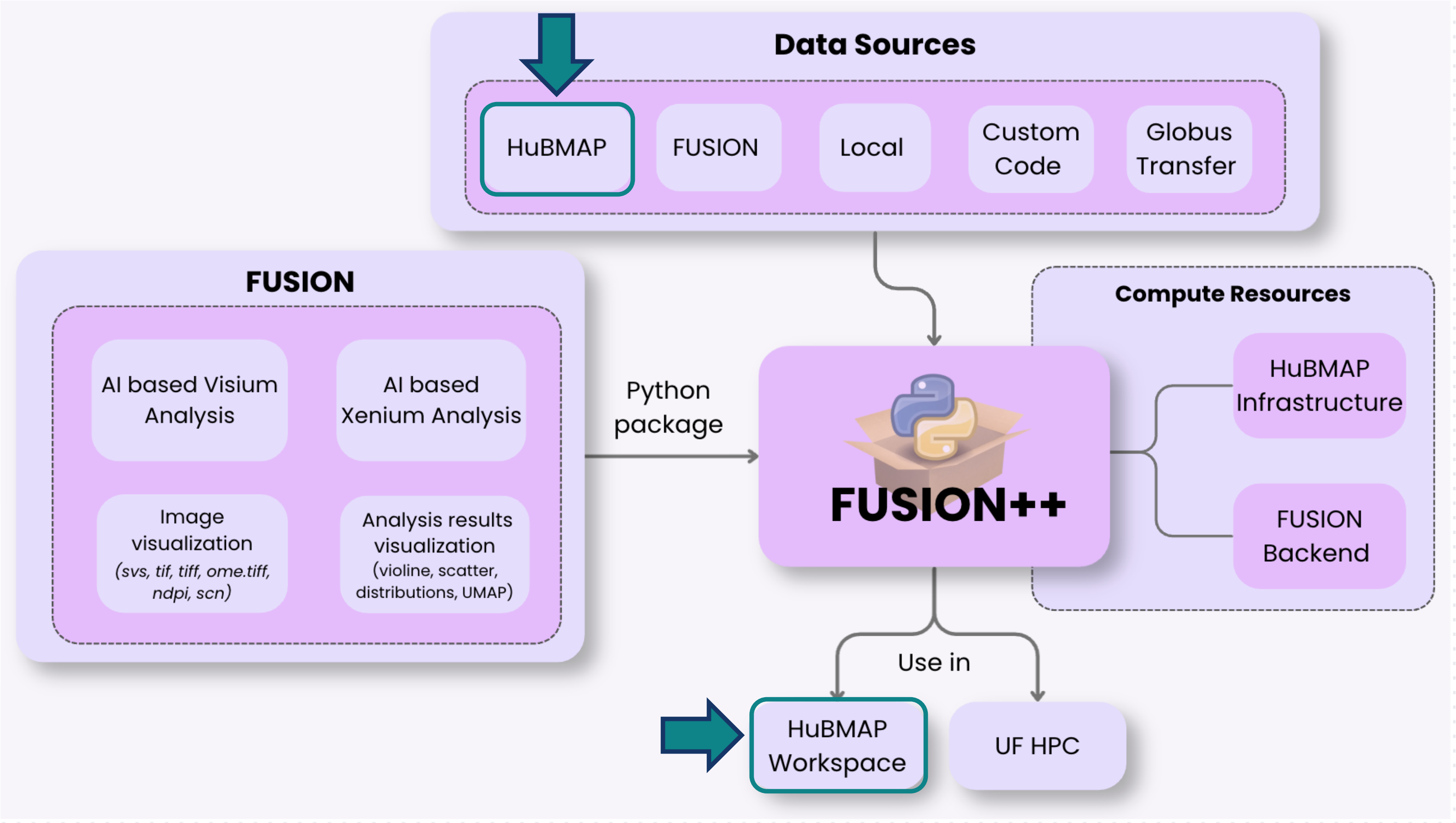
Compare distributions across conditions reveals a continuous injury spectrum across FTUs, beyond a single histological score

Plot → Slide Lasso

Circle outlier points in a plot; the viewer instantly pans and zooms to those exact structures on the WSI

Live Fusion Walkthrough via Fusion JupyterHub Interface

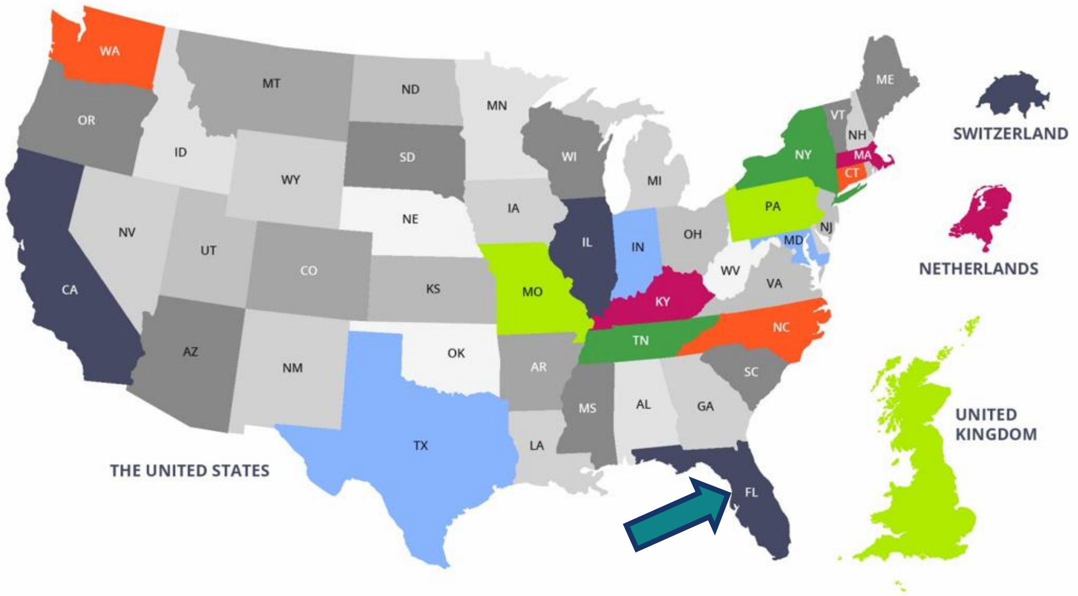
FUSION++ Architecture



* Xenium pipeline integration with FUSION++ in progress

HuBMAP Consortium

Human BioMolecular Atlas Program

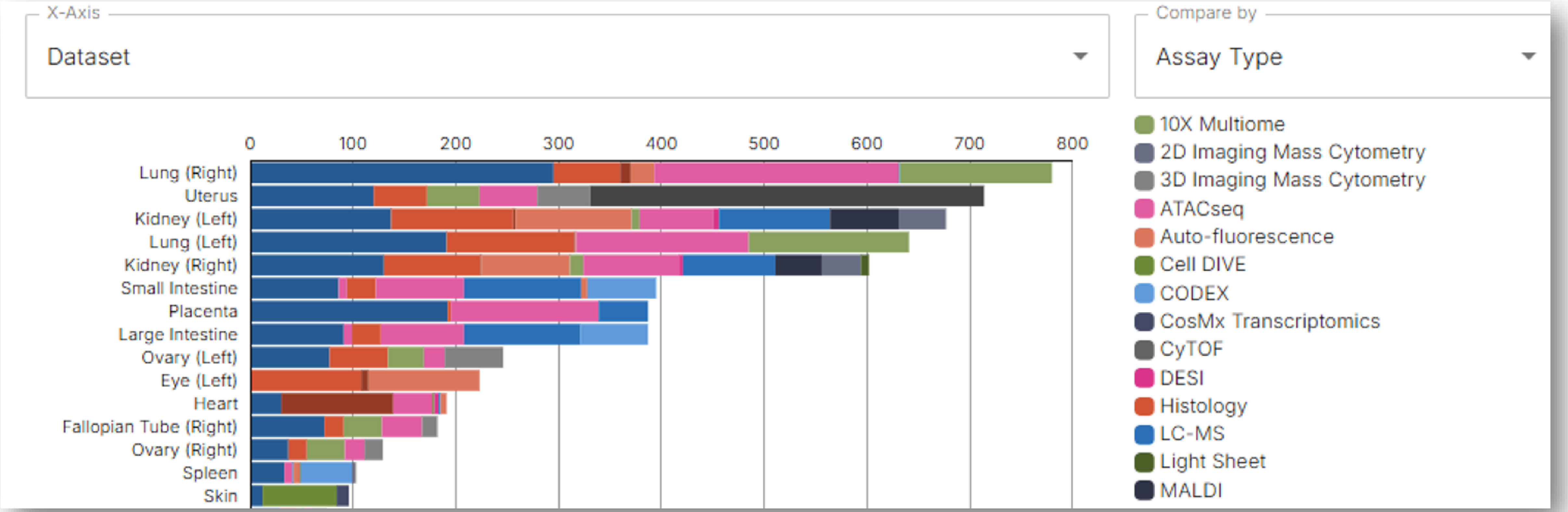


- Create an open, global atlas of the human body at the cellular level
- Develop tools, resources, and cell atlases to understand cell cel cell relationships impact on human health

42 Contributing Sites | 15 States | 4 Countries | 60 Institutes | 400 Researchers

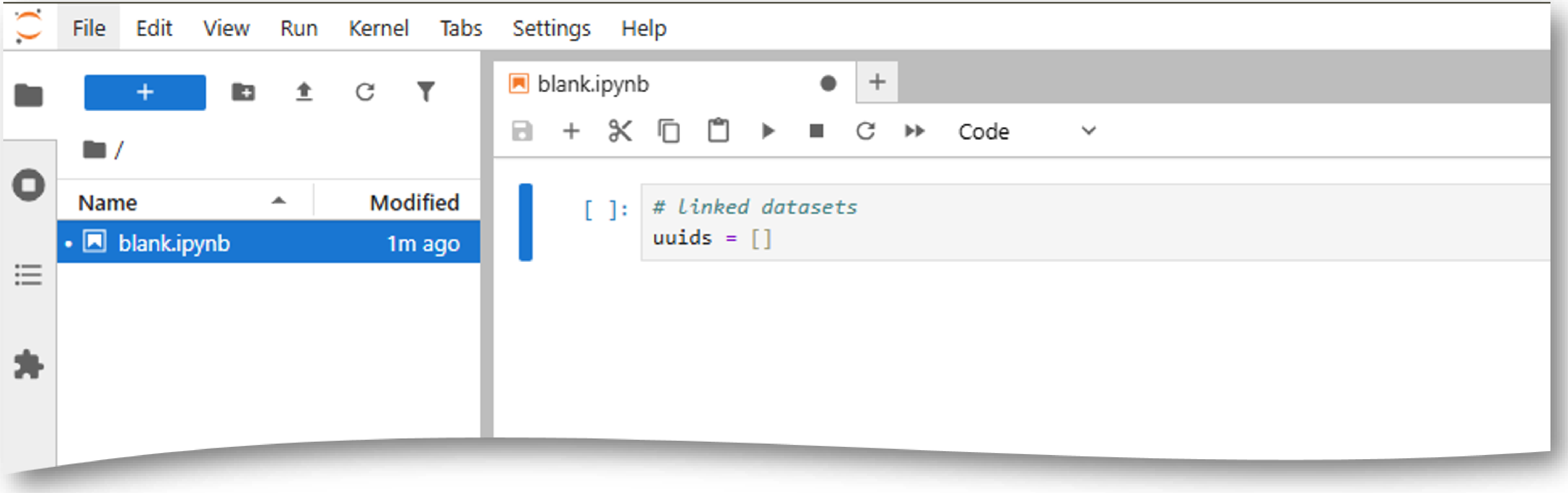
HuBMAP Data

6,093 Datasets | 31 organs | 20+ assays | 432 donors

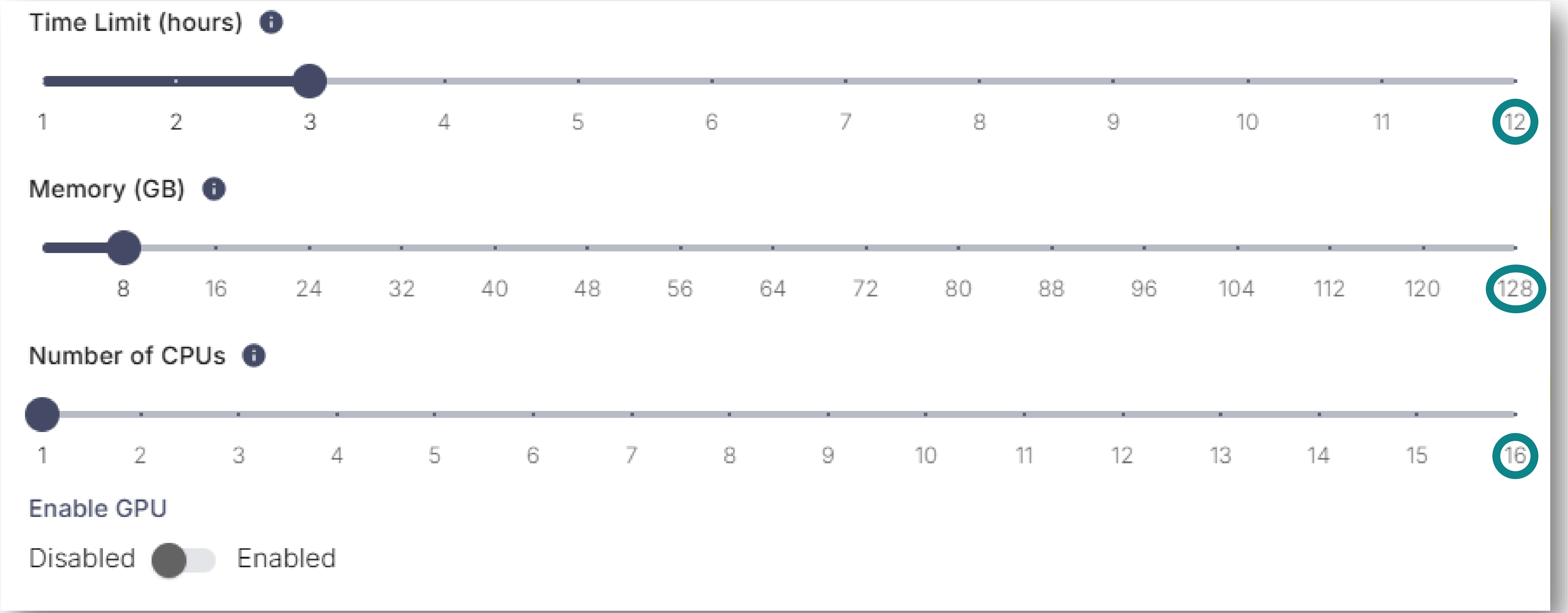


HuBMAP Workspace

- A lightweight exploration platform (jupyterHub)
- Access HuBMAP data and perform analyses directly within the portal.



HuBMAP Resources

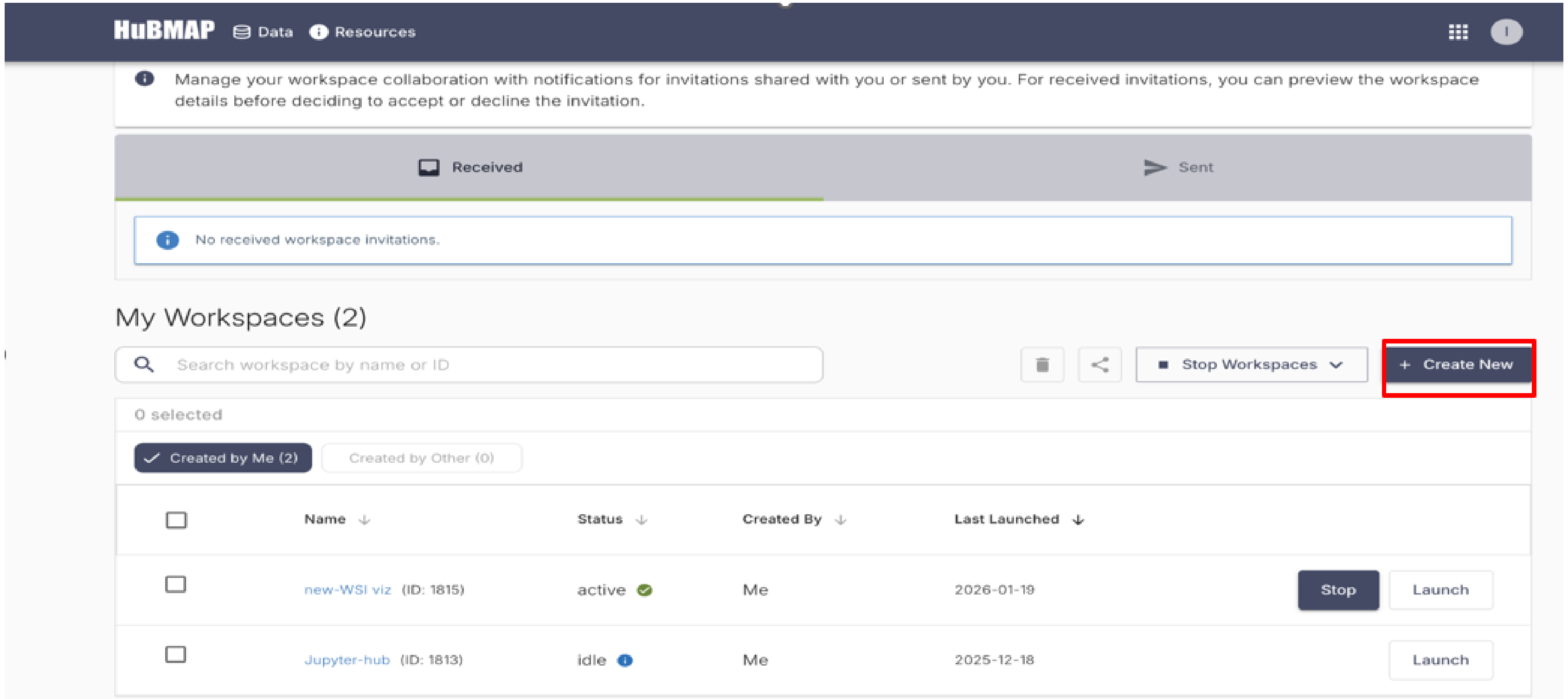


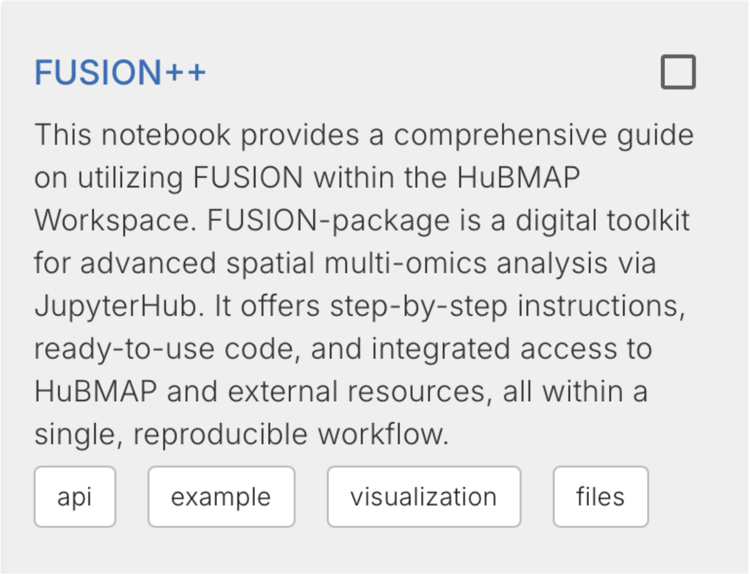
*Per individual

Live Fusion Walkthrough via Fusion JupyterHub Interface

HuBMAP Workspace Access

Prerequisite: HuBMAP workspace access – https://hubmapconsortium.org/workspaces-sign-up-9362/





What's next for FUSION (2.0)

- Integration of single-cell spatial omics (Xenium in-situ).
- Plugin based differential expression, cell neighborhood and niche analysis.
- Agentic AI ecosystem from data ingestion to molecular discovery.
- Broader HPC support beyond workspace (inside institutional boundary).

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